James Irwin

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Security Clearance: Active Top Secret

Education

Cal Poly, San Luis Obispo | MS in Computer Science — Expected: May 2027

- **Proposed Thesis:** *Understanding the Design and Impact of Digital Interfaces for Music Creation*
- **GPA:** 4.00/4.00

Cal Poly, San Luis Obispo | BS in Computer Science — Graduated: June 2026

- **Minors:** Data Science and Philosophy
- **GPA:** 3.76/4.00
- **Award:** Cal Poly Merit Scholar

Skills

Languages: C, C++, ChucK, MATLAB, Python, R, SQL

Tools & Frameworks: Azure, Flask, Git, NumPy, pandas, scikit-learn, Unity

Work and Research Experience

Software Engineer Intern | Northrop Grumman – San Diego, CA — June 2026 – Present

- Worked full-time during Summer 2026 and have continued part-time while completing master's degree.
- Developing an application to process and visualize 3D flight data in real time.
- Designed, implemented, and tested an end-to-end system for transmitting RTK corrections via NTRIP between a base station and a moving receiver.

Machine Learning Research Assistant | Sound Ethics – Remote — Jan 2026 – June 2026

- Collaborated with Sound Ethics through Cal Poly's Data Science Capstone to research methods of giving artists credit through controllable AI music generation.
- Developed a multi-input music generation pipeline using ACE-Step, Demucs, and librosa that enabled conditioning on instrument components from up to five reference tracks.
- Conducted a 14-participant blind user study comparing our framework with baseline ACE-Step. Our framework received higher average ratings for sound quality, prompt adherence, input adherence, and cohesiveness without training a new model.
- Presented research findings at the UCSB Data Science Capstone Showcase.

Software Engineer Intern | Speridian Technologies - Remote — June 2025 – Sept 2025

- Researched and evaluated open-source alternatives to Azure AI Document Intelligence for automated PDF processing.
- Developed Python pipelines to transform OCR model output from PDFs into structured JSON.
- Built a document classification tool in Azure that generated vector representations and measured similarity between documents.

Projects

SoundLink — Winter 2026 - Spring 2026

- Developed a six-level music-based puzzle game in Unity where players arrange 2D puzzle pieces to construct rhythmic compositions through visual and auditory cues.
- Integrated ChucK with Unity to synchronize audio timing with game events.
- Conducted playtesting and improved game based on user feedback.

Photographer Attribution with Neural Networks — Fall 2025

- Built a photographer-attribution neural network in TensorFlow using MobileNetV2 to classify images from a 505-photo dataset without relying on EXIF metadata.
- Created a feature pipeline that combined RGB and HSV image embeddings, composition features, and 20 hand-labeled subject features.
- Used 5-fold stratified cross-validation and balanced accuracy for hyperparameter tuning. Improved cross-validation performance from 79.4% to 87.8% balanced accuracy.